

Week 3. Class 6. Biocompatible materials in Engineering Design

Biocompatibility n. /ˌbaɪəʊskəm.pəˈtəːbɪləti/ (биосовместимость)	How well a material performs in the body without causing unacceptable harm.	Biocompatibility is the first requirement for any implant material.
Adverse reactions n. pl. /ˈædvɜːs riˈækʃənz/ (неблагоприятные реакции)	Harmful or unwanted responses of the body to a material or device.	Adverse reactions may occur even when the device works mechanically.
Inflammation n. /ˌɪnfləˈmeɪʃən/ (воспаление)	A biological reaction (often swelling, redness, pain) that can occur when the immune system is activated.	Inflammation around an implant can damage surrounding tissue.
Rejection n. /rɪˈdʒekʃən/ (отторжение)	A strong immune response that attacks an implanted material or device.	In severe cases, rejection can lead to device failure.
Immunogenicity n. /ˌɪmjʊ.nəˈdʒenɪsɪti/ (иммуногенность)	The likelihood that a material will trigger an immune response.	Engineers test immunogenicity to reduce long-term risk.
Bioactive coating n. /ˌbaɪəʊˈæktɪv ˈkoʊtɪŋ/ (биоактивное покрытие)	A surface layer that is designed to actively improve biological response (e.g., support integration).	A bioactive coating can help tissue integrate with an implant.
Leach v. /liːtʃ/ (выделять вещества в окружающую среду)	To release substances from a material into surrounding fluids/tissues over time.	Some polymers can leach chemicals into body fluids.
Leaching n. /ˈliːtʃɪŋ/ (вымывание, выделение веществ)	The process of releasing substances from a material into surrounding fluids/tissues over time.	Leaching can increase toxicity risk if harmful compounds are released.
Contamination n. /kənˈtæməˈneɪʃən/ (загрязнение)	Unwanted substances introduced during manufacturing/processing that can create safety risks.	Strict protocols reduce contamination during production.

Microbial colonization <i>n.</i> /maɪ'kroʊbiəl 'ka:lənə'zeɪʃən/ (микробная колонизация)	When microorganisms like bacteria attach to and grow on a material's surface.	Microbial colonization increases infection risk after surgery.
Antimicrobial coatings <i>n. pl.</i> /,æntaɪmaɪ 'kroʊbiəl 'koʊtɪŋz/ (антимикробные покрытия)	Surface layers that kill bacteria or prevent bacterial growth.	Antimicrobial coatings are used to lower infection rates.
Hydrophilicity n. /haɪdʀoʃfi'lisiti/ (гидрофильность)	Water-attracting surface behavior that often influences how cells and bacteria interact with a material.	Hydrophilicity can affect protein adsorption on a surface.
Scaffold n. /'skæfəld/ (каркас для роста тканей)	A temporary structure that supports cells and guides tissue growth in tissue engineering.	A scaffold provides support while new tissue forms.
Porosity n. /pɔ:'ra:siti/ (пористость)	The presence of pores/voids in a material, which can support tissue ingrowth or drug release.	Porosity must be tuned to balance strength and transport.
Autoclaving n. /'ɔ:toʊ 'kleɪvɪŋ/ (автоклавирование)	Steam sterilization at high temperature and pressure.	Autoclaving is effective but may damage some polymers.
Ethylene oxide (EtO) <i>n.</i> /'eθəli:n 'a:ksaɪd/ /ɪ: ti: 'oʊ/ (оксид этилена)	Gas sterilization method often used for heat-sensitive materials.	Ethylene oxide is used when heat would deform the device.
Total hip replacement <i>n.</i> /toʊtəl hɪp rɪ 'pleɪsmənt/ (полное эндо- протезирование тазобедренного сустава)	A common medical implant that replaces the hip joint to restore function and reduce pain.	A total hip replacement can restore mobility after joint damage.

Regenerative biomaterials <i>n. pl.</i> /rɪˈdʒenərətɪv ˌbaɪoʊməˈtɪriəlz/ (регенеративные биоматериалы)	Biomaterials engineered to help the body regenerate tissue, ideally back to its original form.	Regenerative biomaterials support healing rather than only replacement.
Biodegrade <i>v.</i> /ˌbaɪoʊdɪˈɡreɪd/ (биологически разлагаться)	To break down gradually inside the body over time.	Some scaffolds biodegrade after they guide tissue growth.
Biodegradation <i>n.</i> /ˌbaɪoʊˌdeɪɡrəˈdeɪʃən/ (биоразложение)	The gradual breakdown of a material inside the body over time.	Biodegradation rate should match the pace of tissue repair.
Injected <i>adj.</i> /ɪnˈdʒektɪd/ (введённый инъекцией)	Delivered into the body using a needle/syringe.	The nanoparticles are injected to reach the target tissue.
Blood–brain barrier <i>n.</i> /ˌblʌd ˈbreɪn ˈbæriər/ (гематоэнцефалический барьер)	A protective barrier that blocks many substances from entering the brain from the bloodstream.	Many drugs cannot cross the blood–brain barrier .
Bioglass <i>n.</i> /ˌbaɪoʊˈɡlæs/ (биостекло)	A glass-like biomaterial that can dissolve in the body, bond to bone, and stimulate new bone growth.	Bioglass can stimulate bone regeneration through released ions.
Dissolve <i>v.</i> /dɪˈzɑːlv/ (растворяться)	To gradually break apart and mix into a liquid in the body; the material can dissolve over time.	Over time, the material may dissolve in body fluid.
Bond to bone <i>v. phrase</i> /bɑːnd tə boʊn/ (соединяться с костной тканью)	To attach strongly to bone tissue, forming a stable interface.	Some coatings help an implant bond to bone more effectively.
Sol–gel process <i>n.</i> /ˌsɔːl ˈdʒel ˈpraːses/ (золь-гель процесс)	A chemistry-based method of making glass-like materials by building a network through chemical assembly.	The sol–gel process can produce bioactive glass structures.

Brittle <i>adj.</i> /'brɪtəl/ (хрупкий)	Hard but easy to crack or break.	Glass can be brittle under repeated stress.
Synergy <i>n.</i> /'smɜːrdʒi/ (синергия)	A combined effect where two materials together perform better than each one alone.	A polymer–glass composite may show synergy in properties.
Surface chemistry <i>n.</i> /'sɜːfɪs 'kɛmɪstri/ (химия поверхности)	The chemical properties of a material's surface that affect how cells interact with it.	Surface chemistry influences cell attachment and growth.
Biosensing <i>n.</i> /'baɪəʊ ,sensɪŋ/ (биосенсинг, обнаружение биологических сигналов)	Detecting biological signals (e.g., markers of disease or pathogens) using devices/materials.	Biosensing can help detect biomarkers in body fluids.
Biosensors <i>n. pl.</i> /'baɪəʊ ,sensərz/ (биосенсоры)	Devices/materials used to detect biological signals (e.g., markers of disease or pathogens).	Biosensors are used to identify disease markers quickly.
Surgical intervention <i>n.</i> /'sɜːdʒɪkəl ,ɪntər 'venʃən/ (хирургическое вмешательство)	A surgery required when the body cannot heal/regenerate a defect on its own.	Early treatment can sometimes reduce the need for surgical intervention.